

Computer Engineering Department

King Mongkut's University of Technology Thonburi

CPE 241 Database

Problem Session #08 Sub-queries

Date: Check PS08 assessment activities on LEB2

Introduction

In this lab, you will design and implement three SQL queries that use subqueries. These queries should demonstrate complex data retrieval techniques, such as filtering results, performing calculations, or aggregating data based on nested conditions. This will enhance your ability to write efficient and structured SQL queries while reinforcing your understanding of relational database operations.

Task List

- **Design Additional Reports for Admin Users and Develop Backend Queries**

Objectives

1. To design report structures involving subqueries.
2. To develop advanced SQL queries using subqueries.

Instructions

STEP 1: Identify three meaningful use cases where subqueries would be useful.

Examples of subquery use cases:

- Filtering results based on conditions from another query
- Aggregating data before using it in a main query
- Comparing values between tables dynamically

STEP 2: Prepare report documentation

For each report, you must clearly describe

- The business question it answers.
- The tables involved and the attributes used.
- Create visual outputs for all designed reports with mockup data. These may include structured tables or charts. Visualizations should clearly reflect the analytical objective of each report and be properly labeled (title, axis labels, units, legends if necessary).

STEP 3: Query Implementation

- For each designed report, write the corresponding SQL query.
- Use clear indentation and readable SQL formatting.
- Provide comments explaining complex logic.
- Example queries include:

STEP 4: Query Validation

- Test each query with sample data.
- Ensure the results match the intended business question.
- Verify correctness of subquery results.

Deliverables

1. Three finalized SQL subqueries.
 2. All SQL queries used to generate each report.
 3. Screenshots or exported results of query outputs.
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Final Submission

- Compile all written explanations and pictures into a single a PDF file and ensure it is named appropriately (e.g., **PS08_YourGroupName . pdf**) then submit the report on LEB2 before the deadline.
- Upload the PDF report to the Assessment Activities section for PS08 on LEB2.